Silas Weir Mitchell (1829–1914): the rewards of versatility and its price

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Summary: Dr Silas Weir Mitchell, known as the father of neurology in the United States and as the author of several best-selling novels, pursued many interests during a long career in medicine and literature. He was also a renowned public figure and an able laboratory scientist who, according to President Theodore Roosevelt, served his country well. Mitchell was well ahead of his time and his refusal to conform to certain rigid ideas about what was and what was not permitted to a physician brought him into conflict with some of his contemporaries. This paper examines aspects of Mitchell's versatility, the rewards it brought him and the price he paid for it.

In his presidential address to the sixth congress of American Physicians and Surgeons, held in Washington, DC, on 12 May 1903, Dr William W Keen referred to Dr Silas Weir Mitchell (1829–1914) (Figure 1) as "the most brilliant medical genius that America has produced within my personal recollection"[1]. Remembered mainly as the father of American neurology, as a gifted and sometimes best-selling novelist, and as the discoverer of a rest treatment for nervous exhaustion that made him world famous, Mitchell came in for more than any one man's share of praise during his lifetime. Moreover, he was noted for "his charm of manner, his patient and studious search for truth, and his philosophic view of life in all of its vicissitudes and potentialities" and had "an extraordinary ability to radiate [these qualities]... to others and to infuse inspiration and knowledge into those who were associated with him. No man could work under him without becoming not only a better doctor, but also a better man"[1]. President Theodore Roosevelt

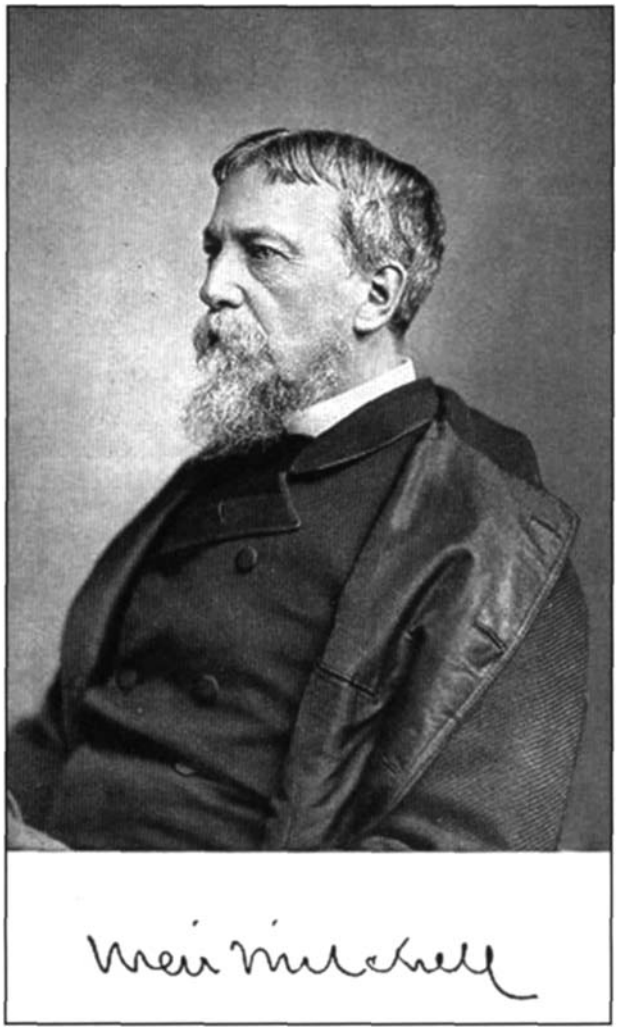

Figure 1. Silas Weir Mitchell. (Courtesy of the Clendening History of Medicine Library, University of Kansas Medical Center, http://clendening/kumc.edu/dc.)

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referred to Mitchell as one who served his country well and there were literary critics who held that his historical fiction was superior to that of William Makepeace Thackeray (1811–1863)[2].

The sum total of these many compliments comprises some, but not all, of the rewards of a many-faceted career. The price paid for it was not small and included accusations of charlatanry, popularization and quackery[2,3]. How could a person who was held up as an example in medicine, literature and public life have been referred to in such terms? Were such slurs no more than expressions of pettiness and envy on the part of those who uttered them, or was Mitchell so far ahead of his time and so versatile that he was able to achieve things which the common wisdom of his day held to be incompatible with a career in medicine? In this article we seek at least partial answers to these questions in Mitchell's writings and in those of his contemporaries.

Literature and medicine

A biographer of half a century ago wrote that "Philadelphia has always regarded artists as queer fellows"[4]. This statement suggests that Mitchell's colleagues may have found his twin careers as novelist and physician disturbing to their idea of how medicine should be practised. An obituary for Sir Benjamin Ward Richardson (1828–1896), an English contemporary of Mitchell's who was also something of a polymath, pictures him as:

> one of those happy men who know how to blend their tastes with their interests, and how to show the public that they possess that knowledge. In the medical profession this task is especially difficult, as people have an idea that a doctor should do nothing but doctoring, and that if he goes about lecturing and takes up a cause he probably has little or no practice.[5]

Richardson wrote poetry and worked in his laboratory as a "bench scientist" while also practising general medicine and pulmonology[6]. The burden of proof, it would seem, was on the doctor to show his patients that his blending of tastes and interests would not be to their detriment.

That Mitchell recognized the need to prove himself if he wished to indulge in literature while practising medicine is indicated by the following statement of his: "Feeling that I could not be injured by literary success, and trusting the good sense of the American people to know whether I was any the less a good doctor because I could write a novel, I continued to thus amuse myself"[2]. John Greenleaf Whittier (1807–1892), the well known poet and abolitionist with whom Mitchell corresponded, once wrote to him: "[Your poems] prove as in Dr [Oliver Wendell]... Holmes's case, that science and medical skill are not incompatible with the true poetic feeling and rhythmic felicity"[2]. The words "not incompatible" imply that, in the popular mind at least, poetry and medicine did not go together.

Although Oliver Wendell Holmes (1809–1904), originally a general practitioner and later professor of anatomy and physiology at Dartmouth and Harvard, as well as a poet and essayist of note, was Mitchell's senior by 20 years; the two were fast friends[7]. "The younger man often sought refuge in the sympathy of Boston, when his colleagues in Philadelphia shook their heads over the idea of a career uniting medicine with literature"[2]. On one occasion Holmes wrote to him: "If you go on writing such charming verse, you will spoil yourself for a doctor, for nobody will believe you can do two things as well as you can do this one thing"[2].

Thus we discern, in Mitchell's own writings, in those of his contemporaries, and even in comments made abroad and not pertaining directly to him, a widely held and popular opinion that medicine requires all of a practitioner's time and does not mix with other kinds of activity. People defying it, as Mitchell did on more than one score, would certainly find the quality of their medical practice held suspect.

Science and medicine

In this day of clinician scientists and university tracks combining a PhD with a medical degree, it is difficult to grasp that, in Mitchell's time, some physicians regarded even scientific research as an illegitimate field of endeavour for a doctor. When he still entertained hopes of being appointed to a chair in physiology at one of the two great teaching centres in Philadelphia, Mitchell spent much of his time investigating such subjects as snake venom, the functioning of the cerebellum and the antagonism between morphine and atropine. These activities were looked at askance by older colleagues and not only in his home city – Mitchell reported that:

> The late Professor Henry Smith said to me, "What nonsense to bother yourself about snake poisons"! And a larger man, Professor Samuel Jackson, warned me that every experiment in the laboratory would lose me a patient. I did not think so.[2]

In 1867 Holmes, in an introductory lecture to the medical class at Harvard, stated the anti-science position bluntly:

> We all admire the achievements of... [the] distinguished band of doctors who do not practice. But we say of their work and of all pure science... it is very splendid, but it is not a practising doctor's business. His patient has a right to the cream of his life and not merely to the thin milk that is left after "science" has been skimmed off. The best a physician can give is never too good for the patient.[8]

The position Holmes stated is lent some support by Mitchell himself. He had done invaluable service at Turner's Lane Hospital in Philadelphia during the Civil War and, with Drs Keen and Morehouse, had taken some 2000 pages of notes on amputees and other soldiers with peripheral nerve

injuries. The result was a book entitled *Gunshot Wounds and Other Injuries of Nerves*[9], concerning the writing of which Mitchell stated: "I so much dread to find increasing practice or other cause removing me from the time or power to search for the new truths that lie about me so thick"[2]. He was, then, acutely aware of the clash between research and medical practice; the latter ultimately won because he was burdened with dependants.

Still, there is no overestimating the importance of the work done by Mitchell and his colleagues. A full century later, a historian describing the achievements of the staff at Turner's Lane said: "Yet here, following the precepts of Hippocrates and Sydenham, there was written a chapter in medical history during the Civil War that has affected clinical thought to the present day"[10]. However, the claim this work made on Mitchell's time and what, according to Holmes, was due his patients was large.

Health promotion and the rest treatment

Mitchell's book *Wear and Tear*, published in 1871, may be regarded as an early attempt at health promotion[11]. It enumerated the ills arising from the American propensity to take life too seriously and from an unwillingness to play, the upshot being a rising incidence of what was then referred to as nervous disease. Today we would speak of neuroses but, in Mitchell's day, their presentation was often neurological and so they came within the purview of neurology rather than of psychiatry[12].

The book's emphasis was on prevention, a controversial subject in the United States during the last quarter of the nineteenth century. There was even some suspicion that it represented a kind of self-advertisement:

> Physicians... for the most part then resented any efforts to enlighten the public in regard to medical matters. In those years the medical guild still spoke its own private language and still maintained that its labours should concern themselves with disease rather than the conditions which produce disease.... The book... therefore... struck a new note and caused a great deal of unfavourable comment from the more old-fashioned among his colleagues. The deadly word "popular" was spoken, later it was superseded by "charlatan".[2]

At the time of the book's publication, some wags referred to it as *Weir and Tear*.

Closely connected with *Wear and Tear* was another book, which appeared in 1877, *Fat and Blood and How to Make Them*[13]. It may be regarded as a psychiatric treatise[14], as it outlines the treatment of nervous exhaustion, or hysteria, by means of rest, overfeeding, electrical stimulation and massage. It was translated into many languages and Freud is said to have adapted it for use in his practice[14]. It made Mitchell world famous and, despite the fact that his forceful personality, suggestion, tact and knowledge of human nature had much to do with what the method achieved, it worked for many others too.

In connection with it there were, once again, murmurings of charlatanism[15]. In 1883 W S Playfair (1835–1903), Professor of Obstetric Medicine at King's College, London, wrote a monograph entitled *The Systematic Treatment of Nerve Prostration and Hysteria*[3]. It was written to further the cause of Mitchell's "rest cure" and to attest to its successful employment in hands other than those of a highly skilled neurologist. In the sixth edition of *Fat and Blood*[16] Mitchell quoted Playfair extensively, while the latter, in his own little book, stated: "One or two of my professional brethren, whose opinion I value highly, have objected to the methods of treatment employed, especially to the massage, as savouring, to speak plainly, of quackery"[3]. Here we have first-hand testimony of the kind of accusation levelled at Mitchell.

Renaissance man

It is instructive to review briefly the many professions Mitchell followed, according to his various biographers. One refers to his "da Vinci-like versatility" and tells of his skill as a mechanic[1]. He had a room in the old Orthopaedic Hospital in Philadelphia that "was full of apparatus he had constructed for the measurement of tremors, the testing of muscle tension and of reflex responses, and for the study of gaits"[1]. Another biographer calls him a toxicologist, as her focus is on his work with snake venom and his introduction of an entirely new concept to the field: that more than one poison may be involved in any given case of envenomation[15]. He is spoken of, too, as advancing our knowledge of pharmacology by his studies on ether, atropine, morphine and cannabis indica[1]. As a physiologist he investigated the functioning of the cerebellum and the effect of cold on the skin of birds, when he worked for a time in Paris in the laboratory of Claude Bernard (1813–1878)[2]. There were many who spoke of him as a psychiatrist, some going so far as to claim that he was the greatest in America during the last third of the nineteenth century[17]. He is considered the founder of neurology in the United States[1] and his preeminence in that area needs no further documentation. He was a novelist, a poet and a writer of books for children[15], as well as the author of over 250 medical papers, most of them composed when he was engaged in full-time practice.

It is suggested that Mitchell engaged in politics, when he helped to break the "boss system" in Philadelphia; he was, moreover, consulted on matters of state by two US Presidents[1]. Underlying all of these activities and, to a certain extent, serving as a common denominator among them was a kind of impassioned pragmatism. Although he was spoken of as possessing flare and panache, neither of these was in evidence

when it came to proving a scientific point[2]. In an address delivered on the fiftieth anniversary of the American Medico-Psychological Association, in 1894, he accused the assembled psychiatrists of neglecting, much to their detriment, teaching, careful observation and meticulous record-keeping in their institutions[18]. He recognized that the presence of young, questioning minds was essential to the proper functioning of any medical service, especially if it had pretensions to academic status. Finally, his instruction extended beyond youthful physicians to the nursing staff, thus prefiguring the modern concept of teamwork. The neuropsychiatric sisters he trained, who became known as Weir Mitchell nurses, often had no hospital experience whatever[1].

Success

Thus far, it would seem that the calumnies against Mitchell arose from his being far in advance of the day. It was expected of him, as of any other physician, that he would devote himself entirely to his practice. Instead, he worked as a physiologist in his spare time and, from the age of 40, wrote novels – without diminishing, by so much as an iota, his commitment to medicine. Indeed, he functioned both as a general practitioner and as a consultant neurologist at the Philadelphia Orthopaedic Hospital and Infirmary for Diseases of the Nervous System[1]. At a time when health promotion was frowned upon, to the point of being alluded to as advertising charlatanism[19], unless presented in the name of some authorized body, alone he wrote a book on the subject. Finally, his proposed treatment for nervous exhaustion, which, admittedly, made use of the as yet unnamed placebo effect, was regarded with mistrust, possibly because it brought him such fame. Thus Mitchell's penchant for swimming against the stream led to much resentment: "[He] was a complex, highly intelligent man who followed his own judgement no matter how unfashionable the path; his rise to medical and literary fame was not without its vicissitudes"[15].

A further spur to the calumnies against Mitchell was quite simply his astounding success at everything to which he set his hand in the field of the intellect. His novel *Hugh Wynne* sold over 500,000 copies and was ranked with *The Scarlet Letter* as America's greatest work of fiction[2]. One citation for an honorary degree went so far as to call him "the chief ornament of medical science in the New World"[2].

A disease, erythromelalgia, then considered a "vaso-motor neurosis" but since associated with thrombocythaemia, was named after him[20] and he has been accorded primacy in other delineations[21,22]. Many of his scientific predictions came true, in particular one made to Dr Simon Flexner (1863–1946) that the organism supposed to be responsible for poliomyelitis would require much more powerful microscopes for its demonstration[1]. He did work on the sturgeon's blood crystals that led to advances in understanding the crystallography of haemoglobin 50 years later[23] and he foresaw psychiatry's present preoccupation with the chemical manipulation of mood[24]. Finally, he predicted that the wider availability of mescal would lead to a wave of addiction[25].

Mitchell's standing with the most distinguished of his contemporaries was high. Neurologists including Keen, Brown-Séquard and Charcot spoke of him in glowing terms[1,26,27]. He was awarded honorary degrees by some of the Western world's finest institutions of higher learning[2] and by the Royal Society, of which he was made a Foreign Fellow[15]. His efforts as an author were crowned with both material and artistic success[2]. As a public figure, he came to the attention of two Presidents of the United States and enjoyed the status of Philadelphia's foremost citizen[15].

In the field of science, although he did some solid work, he was frustrated by his failure to attain a teaching post in physiology, something that might have enabled him to support his dependants without having to maintain a large general practice. He once said that each of his novels contained "a clinic"[2] and one wonders what the fiction would have been like had his patients been restricted to those ordinarily seen by a neurological consultant. Just as his insistence on a man's right to follow his own judgement and lead a multifaceted career brought him the plaudits of success, so it led to calumny and the frustration of his hopes as far as his longed-for career as a laboratory scientist was concerned. It is virtually certain, though, that the triumphs outweighed the disappointments.

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